Modern Academy for Engineering and Technology

Graduation Project Summary Template

Project Name: Smart Dental System Using AI

(Department): Computer Engineering

Supervisor's Name
Dr. Khaled Ahmed Morsy

Participating students: Hassan Mahdi Hassan Diab - Hajar Ahmed Mohamed Fawaz - Reham Mohamed Idris - Reham Mohamed Aboud - Ahmed Osama Mohamed - Israa Mustafa Ali - Sahar Samir Kamal

Project Objectives

The primary aim of this project is to architect and deploy a fully integrated, AI-driven digital dentistry platform. To achieve this, the project is subdivided into the following specific, measurable objectives:

- **Dataset Engineering and Preprocessing:** To acquire, clean, and meticulously preprocess the MOD (Mouth and Oral Disease) dataset, ensuring the color intraoral images are optimized for deep learning models by addressing potential class imbalances and standardizing image resolutions.
- **Multi-Label Pathology Classification:** To design, train, and fine-tune a robust Deep Convolutional Neural Network (utilizing architectures such as ResNet50) capable of performing multi-label classification. This allows the system to simultaneously detect the presence or absence of multiple co-occurring conditions from eight distinct categories with high accuracy and minimal false negatives.
- **Precise Lesion Localization (Object Detection):** To implement a state-of-the-art object detection algorithm (specifically YOLOv8) to not only identify but also visually bound and localize specific conditions within the image, thereby providing visual proof to the user.
- **Intelligent Conversational Agent Integration:** To integrate Large Language Models (LLMs) to interpret the numerical output of the vision models and autonomously generate comprehensive, patient-friendly medical reports. Furthermore, to develop a real-time chatbot that can answer follow-up queries related to the generated report.
- **Cross-Platform Application Development:** To build a robust, scalable backend infrastructure (utilizing Node.js and MongoDB Atlas) paired with intuitive, highly responsive frontend interfaces. This includes a cross-platform mobile application engineered with Flutter and a web portal using the MERN stack, ensuring widespread accessibility.

Pictures from the project



